Improvement proposals and stakeholder feedback

PlaySafe

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1.Introduction

In this document, I explain the feedback collected for my project and the improvements I would like to make in the future.

PlaySafe is an AI model that predicts if a football player could have an injury in the next season, using information from previous seasons.

2. Stakeholder Feedback

Stakeholder: Arantxa de la Chica
Role: Physical Trainer
Club: Olympia Las Rozas

For this part of the project, I asked a real stakeholder to give some feedback of my final model.

"I think PlaySafe is a good idea because injuries are a big problem in football. Knowing which players could have a higher risk of injury could help us pay more attention to them during training.

However, I think the model needs more information about the players. For example, how tired they are, how many times they train per week and how well they recover after matches.

I also think it would be easier to use if it was an application instead of a notebook. It is important to remember that every player is different, so I would use this tool as extra information, but not to make final decisions."

What I Learned from the Feedback

This feedback helped me understand that PlaySafe could be useful for football staff, but it still needs improvements.

The most important points are adding more information about players, making the tool easier to use and remembering that AI predictions should only support the decisions of professionals.

3.Improvement Proposals

Based on the limitations of my project and the possible feedback from a physical trainer, I would like to make the following improvements.

3.1. Add More Data

I would like to add more seasons and more players to my dataset. This could help the model learn from more examples and improve its predictions.

I would also like to include information about fatigue, training intensity and recovery time.

3.2. Improve the Models

In this project, I used Logistic Regression and Random Forest.

In the future, I would like to test different parameters to see if I can improve the results of both models.

3.3. Improve the Train and Test Split

In my current project, I separated the data by seasons.

However, the same player can appear in both training and testing data.

In the future, I would like to test the model with completely different players to see how well it works with players it has never seen before.

3.4. Testing with Real Club Data

I would like to test it with real information from a football club.

This would help me understand if the model works well outside my dataset and if its predictions are useful for football professionals.

3.5. Create a Simple Application

At the moment, the porject works in a Python notebook.

In the future, I would like to create a simple application where coaches and physical trainers can enter player information and see the injury risk.

This would make the project easier to use for people who do not know Python.

4. Conclusion

The feedback example shows that the project could be useful for football professionals, but there are still things to improve.

My main priorities would be adding more information about players, improving the model and creating an easier way to use it.

I think these improvements could make PlaySafe more useful and reliable in the future.